THE LAB — DAY 06

Machines That See

Teaching a machine to see — convolutional neural networks

Day 06 · Week 2 · Unlocks 🕵️ Photo Detective

1 Mission briefing

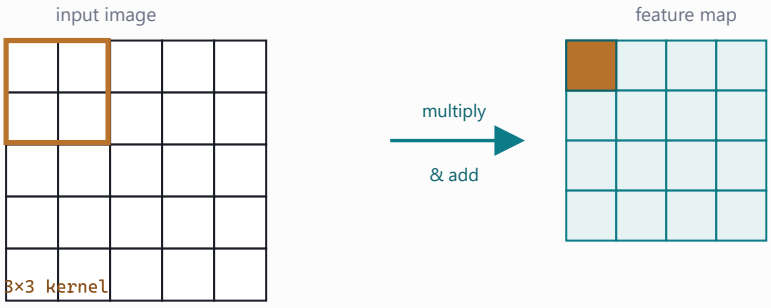
Our lab cameras are technically blind — they record **pixels**, not *things*. Yesterday the Digit Reader flattened each image into one long row and still read tidy handwriting, but real photos have edges, textures, and objects that can sit anywhere in the frame. Today you teach a machine to actually **see** — and turn it loose on 60,000 real photos. *Previously:* you unlocked ✍️ Digit Reader, a neural net that reads handwritten digits.

- ☐ See why **flattening** destroys a photo's structure
- ☐ Slide a **kernel** to find edges, blur, and sharpen
- ☐ Stack convolutions: edges → shapes → objects
- ☐ Train a **CNN** on CIFAR-10 and read accuracy **honestly**
- ☐ Time a step on GPU vs CPU
- ☐ See that a convolution is really a matrix multiply

2 Field guide the concepts

A photo is a grid, not a row. Neighboring pixels belong together — they form an edge, a whisker, a wheel. Flattening throws away "next to". A CNN keeps the grid and studies small neighborhoods at a time.

Convolution is a sliding stencil. Take a tiny 3×3 grid of numbers (a **kernel**), slide it across the image, and at each stop multiply the nine pixels underneath and add them up. One kernel finds edges, another blurs, another sharpens. Each kernel's output is a **feature map**, and stacking layers lets them compose: edges become shapes, shapes become objects.



A 3×3 kernel slides across the image; at each stop it multiplies the nine pixels underneath and adds them into one number of the feature map.

Read accuracy honestly. **CIFAR-10** is 60,000 tiny 32×32 color photos in 10 classes. Around **70%** can feel low — until you remember random guessing is 10%. That is 7× better than chance; humans hit ~94%. A real scientist reports the true number, not the flattering one.

> 🧮 **Matrix Watch —** a convolution is secretly one big matrix multiply: unroll every 3×3 patch into a row (the `im2col` trick) and the whole layer becomes `C = A @ B`. High-performance computing makes that multiply fast.

3 Lab missions check each off in the notebook

1 The vertical-edge kernel ☐

Hand-build a 3×3 edge kernel and slide it with `F.conv2d`.

Expect: vertical outlines light up while flat areas stay dark.

2 A blur kernel ☐

Fill a kernel that averages the 3×3 neighborhood.

Expect: the same photo comes back soft and smeared.

3 Assemble the CNN ☐

Stack `Conv2d` + `ReLU` + `MaxPool` blocks, then `Flatten` into a linear head.

Expect: the feature-map size shrink 32 → 16 → 8 as you pool.

4 Count the cost ☐

Add up the model's trainable parameters.

Expect: a few hundred thousand weights — big, but trainable in class.

5 Build the confusion matrix ☐

Plot `confusion_matrix` of predictions vs truth on the test set.

Expect: a bright diagonal, with cat and dog leaking into each other.

6 Look at a different class ☐

Pick another class and read its row of mistakes.

Expect: the confusions make visual sense (deer ↔ horse, car ↔ truck).

7 Time a training step, GPU vs CPU ☐

Clock one forward+backward step on each device.

Expect: the GPU finish many times faster — a preview of high-performance computing.

4 Cheat sheet convolutions & CNNs

`F.conv2d(x, kernel)`	Slide a kernel over x by hand (no learning).
`nn.Conv2d(3, 32, 3, padding=1)`	Learnable conv: 3 → 32 channels, 3×3, keeps size.
`nn.MaxPool2d(2)`	Halve height & width, keep the strongest signal.
`nn.Flatten()`	Grid → row — only after the conv layers.
`nn.Dropout(0.3)`	Randomly switch off 30% of activations while training.
`Normalize((0.5,)*3, (0.5,)*3)`	Center each color channel to the range [-1, 1].
`confusion_matrix(y_true, y_pred)`	Table of what got confused with what.
`torch.jit.trace(model, ex).save(path)`	Freeze the trained model into one portable file.

5 🔓 Unlock your Studio module

🕵️ Photo Detective

Train the CNN, then trace and save it. The unlock cell must write exactly:

`ai_studio/models/photo_detective.pt` `ai_studio/models/photo_detective_classes.json`

A new tab lights up — upload a photo and watch your detective investigate:

```
$ python ai_studio/app.py
```

6 Mini-glossary

convolution
Sliding a small kernel over an image, multiply-and-add at each spot.

kernel / filter
The little grid of weights that does the sliding.

feature map
The output image a single kernel produces.

pooling
Shrinking a feature map by keeping its strongest values.

stride / padding
How far the kernel jumps · a border that preserves size.

dropout
Randomly switching off neurons in training to fight overfitting.

normalization
Rescaling pixel values so training stays stable.

7 Show & tell

1. Show one photo your Detective nails and one it flubs. What's different about them?
2. Your accuracy vs 10% random vs 94% human — is 70% good? Defend your answer.
3. Which two classes does your confusion matrix mix up most, and does that make sense?